Unit 2, Day 17

An Algorithm with Partial Quotients

Let's make sense of and use an algorithm that uses partial quotients.

Warm-up

Number Talk: Divide by 3

Find the value of each expression mentally.

- $90 \div 3$

- $96 \div 3$

- $960 \div 3$

- $954 \div 3$

Activity 1

Tyler's Method

Tyler uses a different method to find the value of $465 \div 5$. Let's compare Priya's and Tyler's work.

Priya's method

$$\begin{aligned} 400 \div 5 &= 80 \\ 60 \div 5 &= 12 \\ 5 \div 5 &= 1 \\ \hline 465 \div 5 &= 93 \end{aligned}$$

Tyler's method

$$\begin{array}{rl} \boxed{93} & \\ 1 & \\ 12 & \\ 80 & \\ 5\overline{)465} & \\ -\ 400 & 5 \times 80 \\ \hline 65 & \\ -\ 60 & 5 \times 12 \\ \hline 5 & \\ -\ 5 & 5 \times 1 \\ \hline 0 & \end{array}$$

1. How are these methods alike? How are they different? List as many similarities and differences as you can find.

2. Why do you think Tyler uses subtraction in his method?

3. Show how Tyler might record the process of finding the value of $428 \div 4$.

Activity 2

A Stack of Partial Quotients

Jada uses partial quotients to find out how many groups of 7 are in 392.

1. Analyze Jada's steps in the partial-quotients algorithm.

```
  56
   9
   7
  40
7 )392
 − 280
  112
 −  49
   63
 −  63
    0
```

   a. Look at the 3 numbers above 392. What do they represent?

   b. Look at the 3 subtractions below 392. What do they represent?

2. Show another way you can decompose 392 to divide by 7.

3. Use a partial-quotients algorithm to find the value of $702 \div 3$.